

Enterprise engineering studio · Intelligent software for modern commerce

Intelligent systems built to quietly run commerce.

Sanad constructs operational backbones and bespoke autonomous logic for commercial enterprises. We remove the cognitive and administrative weight of scaling — engineering a structural reduction of at least 30% in daily operational load.

[Request Technical Dossier](#)

[Read the 30% baseline](#)



Enterprise data streams route into the Sanad Core modules, then synchronize out to infrastructure the client controls.

[For Leadership](#)

Predictable velocity, margin protection, and operational clarity.

[For Engineering](#)

Decoupled architecture, local-first integrity, and zero black-box dependencies.

01 Proof by metric

The 30% efficiency baseline

Operational load is not a feeling. It is waiting, re-keying and reconciling — three things that can be measured, and removed. Every system we build is specified against one baseline: a structural reduction of at least 30% in daily operational load.

01 Inventory & Supply Sync

Before

18 hours

of multi-channel lag between stock and sales

With Sanad

Sub-second

synchronization across every channel

02 Financial & Reconciled Auditing

Before

Repeated

reconciliations, performed by staff each cycle

With Sanad

Zero-loss

balancing routines, automated end to end

03 Human Error Overhead

Before

Re-keyed

data entered again in each system it touches

With Sanad

Eliminated

structurally: one entry, one record of truth

Definition

$$L = t_{\text{wait}} + t_{\text{rekey}} + t_{\text{reconcile}}$$

daily operational load, in hours

$$L' \leq 0.70 \times L$$

the 30% Law

Daily operational load, indexed baseline = 100

Baseline

100

With Sanad

≤ 70 $\geq 30\%$

02 Systems & architecture

Ready-made modules. Bespoke engineering.

Two explicit offerings, never blended. Pillar A is production-tested software, configured for your business. Pillar B is engineered from scratch, for operational flows that belong to you alone.

Pillar A

A

Turnkey Intelligent Modules

Pre-configured SaaS

High-efficiency, production-tested systems, ready for deployment.

A.1 Inventory & Distributed Warehousing Logic

One stock position across every warehouse, store and channel.

Interface wireframe

SKU	WH-A	WH-B	WH-C	Total
SKU-1042	128	64	0	192
SKU-2210	12	40	33	85
SKU-3087	220	0	18	238
SKU-4410	7	7	7	21
SKU-5002	54	90	31	175

● Synced 0.4 s ago 5 SKUs · 3 nodes

Schema

```
type StockLevel {
  sku      string
  node     NodeId // warehouse or store
  on_hand  int
  reserved int // held for open orders
  version  Lamport // deterministic clock
}
```

Topology n-node, multi-warehouse

Sync event-sourced, deterministic merge

Reservations per-channel holds, auto-expiry

Offline node-local ledger, reconcile on reconnect

Latency < 100 ms in-site, sub-second across channels

A.2 Hospitality & Unit Management Engines

Units, stays and turnover, held to a single source of availability.

Interface wireframe

Units	Housekeeping	This week
101	Mon Tue Wed Thu Fri Sat Sun	
102		
103		
201		
202		

● Occupied ● Turnover □ Blocked no overlaps

Schema

```
type Stay {
  unit      UnitId
  from      date
  to        date
  status    vacant | occupied | turnover
  rate_plan RatePlanId
}
// invariant: no two stays overlap on unit
```

Allocation overlap-free by construction

Housekeeping status transitions create tasks

Channels booking channels kept in step

Rates plans, seasons and restrictions

Handover turnover opens at check-out

A.3 Unified Commerce & POS Ledger Sync

Every sale, online or in store, posts once to one ledger.

Interface wireframe

Sync	Close	Live
POS #A-10441	Account	Dr Cr
24.00	1100 Cash	46.75 -
18.50	4000 Sales	- 42.50
	2200 VAT	- 4.25
VAT 4.25	Σ Dr = Σ Cr	46.75
Total 46.75		

● Balanced, posted once idempotent

Schema

```
type LedgerEntry {
  tx      TxId // idempotency key
  account AccountCode
  debit   decimal
  credit  decimal
  source  pos | web | erp | gateway
}
```

Model double-entry, append-only

Idempotency one key per transaction

Gateways settlement matched automatically

Close daily close with variance report

Audit immutable history, exportable

Pillar B

B

Tailored Commercial Architecture

Bespoke systems

From-scratch systems, engineered for unique, proprietary operational flows.

B.1 Custom internal operating systems

Replacing patchwork spreadsheets

One system of record in place of a folder of spreadsheets: role-based views, enforced workflows, and a history of every change.

record single source of truth

access role-based

history append-only

B.2 Private agentic workflows

Operating securely on client infrastructure

Autonomous logic that runs inside your perimeter, on your hardware or your cloud account. Every action is logged, bounded by explicit permissions, and reversible.

runtime client infrastructure

egress denied by default

oversight approval gates, full action log

B.3 High-availability distributed databases

With offline-first synchronization

Replicated stores that keep accepting writes when the network does not, then converge deterministically when it returns.

writes local-first

merge deterministic

single point of failure none, by design

03 System mechanics & technical rigor

Specified like infrastructure, stated so it can be tested.

For CTOs and technical evaluators

Data ownership Sovereign

Complete sovereignty. Databases synchronize with client-controlled infrastructure.

Reliability Offline-capable

Operational nodes designed to function without internet disruption.

Latency < 100 ms

Sub-100ms internal state changes across multi-terminal setups.

Integrations Deterministic

Clean, deterministic APIs connecting legacy ERPs, payment gateways, and third-party logistics.

Representative event trace

Illustrative sequence, not a benchmark.

```
# multi-terminal sale on the shop floor
t+000 ms term-02 sale.created
t+008 ms node-01 ledger.posted
t+019 ms node-01 stock.reserved
t+037 ms term-01 state.applied
t+041 ms term-03 state.applied
```

```
# network lost: the node keeps working local
t+912 ms term-02 sale.created
t+921 ms node-01 ledger.posted
```

```
# network restored: replicate to client-db
t+4.2 s node-01 replicate
```

Target: state changes under 100 ms across terminals, with no dependency on the network.

The Sanad manifesto

04

Software should clarify the mind, not demand it.

Most enterprise software creates its own bureaucracy. Teams spend hours managing the tool that was supposed to save them time. Sanad is built on the inverse thesis: intelligent software works silently in the background so leaders and builders can dedicate their focus to what actually matters.

Support for the people who lead.

Sanad

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● Accepting new engagements · Remote-first studio

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[Operational blueprints](#)

Shared on request, with the technical dossier.

[Inquiries](#)

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For an architectural walkthrough of any system on this page, write to us directly.

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